HusAero / Evaluation results

HusAero-8n / HusAero-11n

RGB aerial-object detection and single-drone tracking. Recorded evaluation: 6 November 2026. Release v0.3.0-rc5.

HusAero-8n is fine-tuned from Ultralytics YOLOv8n; HusAero-11n is fine-tuned from YOLO11n. The release contributes data curation, training, evaluation and Pi/Hailo integration. It does not introduce a new YOLO architecture.

Measured result	HusAero-8n	HusAero-11n
250-image mAP50-95	72.81%	76.71%
Video follow localization	78.79%	61.30%
File pipeline throughput	56.56 FPS	44.07 FPS

Selected deployment

HusAero-8n is the Pi/video default, selected from validation for tracking quality and speed. HusAero-11n remains an alternative with stronger image-box AP. Both models were tested on Raspberry Pi 5 + Hailo-8.

Reading the results

Image mAP, video localization, confidence and throughput measure different things. Video localization counts correctly localized target-present frames at IoU >= 0.5. It is not a general accuracy percentage. Throughput excludes drawing and video encoding.

The adapted HusAero-8n improved video recall but also produced more output during source-annotated target absence. Misses, duplicate boxes and drift remain possible. Camera capture and pan-tilt control have not been evaluated.

Reproducibility files include model hashes in models/registry.json, saved configurations in configs/, and numeric per-epoch and per-clip records in results/.

01 / Public image test

250 images, five classes, 50 images per class. Identical input image and label hashes for both models. Actual Pi 5 + Hailo-8 inference, 640px, confidence floor 0.001, NMS IoU 0.45 and maximum 300 predictions.

Class / metric	HusAero-8n	HusAero-11n
mAP50	92.18%	93.27%
mAP50-95	72.81%	76.71%
drone AP50-95	40.40%	46.28%
bird AP50-95	78.78%	83.93%
airplane AP50-95	87.82%	90.36%
helicopter AP50-95	76.66%	78.54%
kite AP50-95	80.38%	84.45%

Scoring and scope

AP uses the shared NumPy IoU/101-point scorer in deployment/detection_scores.py. This is not the official COCO evaluator; Ultralytics evaluation may differ. mAP50-95 averages AP across IoU thresholds 0.50 to 0.95. The lower drone AP must be considered alongside the overall mean.

Public starter data

Split	Images	Per class
Train	1,200	240
Validation	250	50
Test	250	50

The 20 sample images (four per class) are part of this test split. Their demonstration results are not independent evidence. The full starter subset is derived from MBAOD v1; it is not the entire private training provenance of the shipped checkpoints.

02 / Recorded-video test

18 clips in two reserved timestamp groups. Each model processed all 17,142 frames: 16,286 source-target-present and 856 source-target-absent. Failures are retained. RGB recorded files run at 20 FPS.

Measurement	HusAero-8n	HusAero-11n
Localized present frames, IoU >= 0.5	12,832 / 16,286	9,983 / 16,286
Follow localization recall	78.79%	61.30%
Raw precision, confidence 0.25	96.67%	97.36%
Raw recall, confidence 0.25	73.90%	52.63%
Follow output during absence	121 / 856	55 / 856
Matched-target ID changes (diagnostic)	221	114
Longest present-frame miss streak	158 frames	714 frames
Decode + inference + follow + logs	56.56 FPS	44.07 FPS

Tracker settings and tradeoff

HusAero-8n: activation 0.10, keep 0.10, max gap 0.25s. HusAero-11n: 0.25, 0.10, 0.15s. Settings were selected on 3,348 validation frames. Follow scores compare configured pipelines; raw detector scores use the same 0.25 threshold.

The previous HusAero-8n control, with the same final tracker/scorer, localized 9,748/16,286 frames (59.86%) and output during 19/856 absence frames. Adaptation raised localization to 78.79% while absence outputs increased to 121/856.

Timing includes recorded-file decoding, Hailo inference, tracking and logs. It excludes rendering/encoding and is not live-camera latency. ID-change counts are diagnostics, not MOT IDF1/HOTA. Source annotations may omit other visible objects.

03 / Training and tracking method

Detector and tracker

RGB frame -> letterbox to 640px -> Hailo YOLO detection -> NMS / nested-drone-box filtering -> one selected-drone track. FlowAssistedTracker combines gated detector association with bounded Lucas-Kanade optical flow during short detection gaps. Ground truth enters only the separate scorer.

Current Pi follow results use this single-target tracker. SORT and ByteTrack were earlier comparison paths. The optional PC demo uses Ultralytics ByteTrack and must not be described as the same tracker as the current Pi follow pipeline.

Adaptation setting	Recorded value
Epochs / image size / batch	10 each / 640px / 16
Training / validation images	2,219 / 721
Optimizer / initial LR / seed	SGD / 0.0003 / 42
Selected checkpoint	HusAero-8n epoch 8; HusAero-11n epoch 6
Augmentation	mosaic 0.2; close_mosaic 3; scale 0.4; translate 0.1
Calibration / compilation	1,024 identical training-only images; DFC 3.34.0, optimization 0

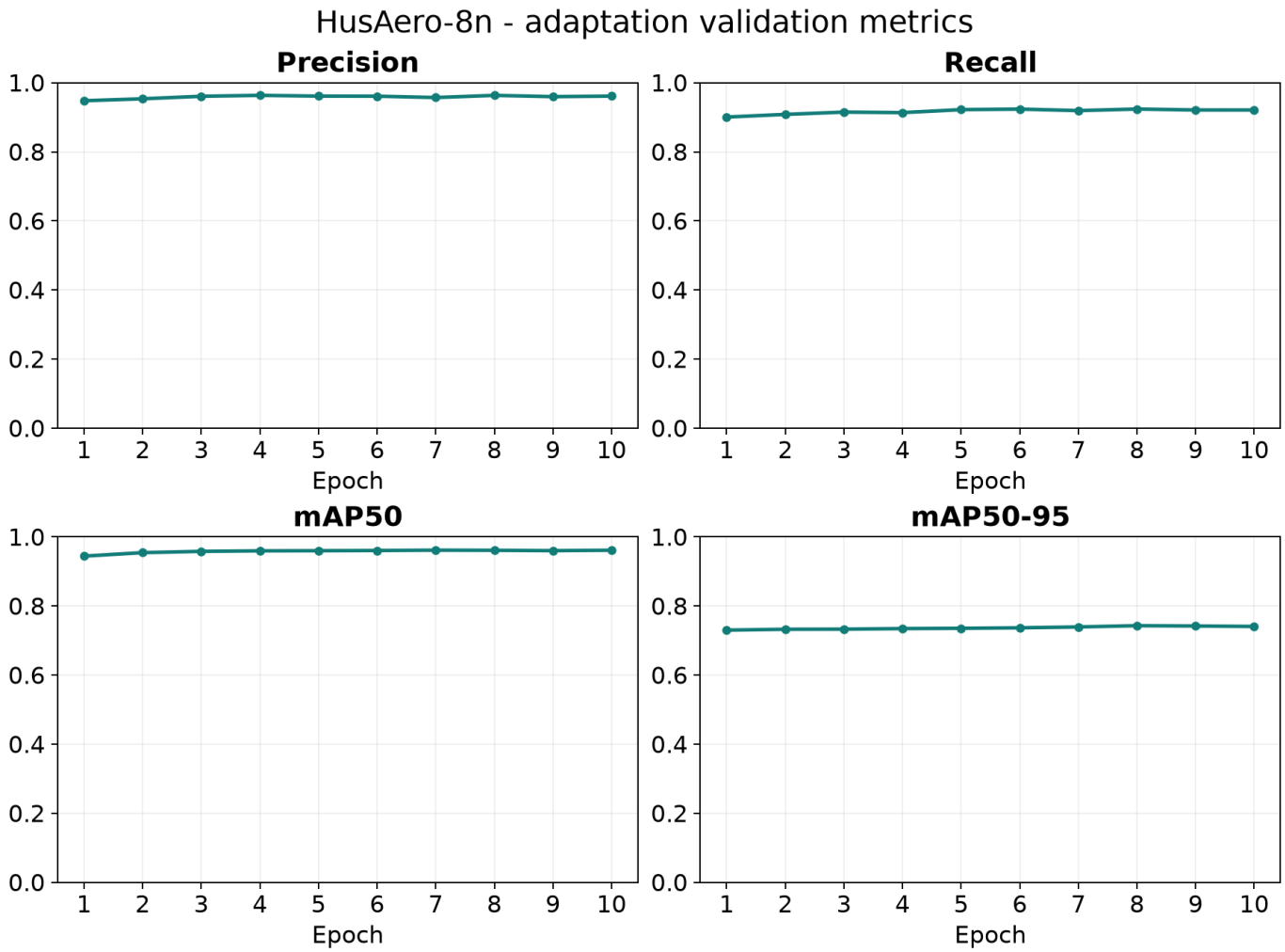
Training history and limitations

The historical matched study completed 50 epochs per model on six classes. Later five-class/rehearsal and adaptation stages have separate CSV histories. The final checkpoints start from different earlier histories; the 10-epoch adaptation comparison does not isolate architecture alone. The separate Pi CPU experiment completed one epoch and is not the deployed checkpoint.

Adaptation training includes 1,200 MBAOD, 353 DUT and 666 video frames; validation includes 250 MBAOD, 200 DUT and 271 video frames. Some original test-folder groups were reassigned to train/validation and are not unseen tests for adapted weights. Reserved groups were excluded from this adaptation training/validation, but baseline first-clip results had been inspected earlier. Parent-training/session independence remains unverified.

04 / HusAero-8n validation metrics

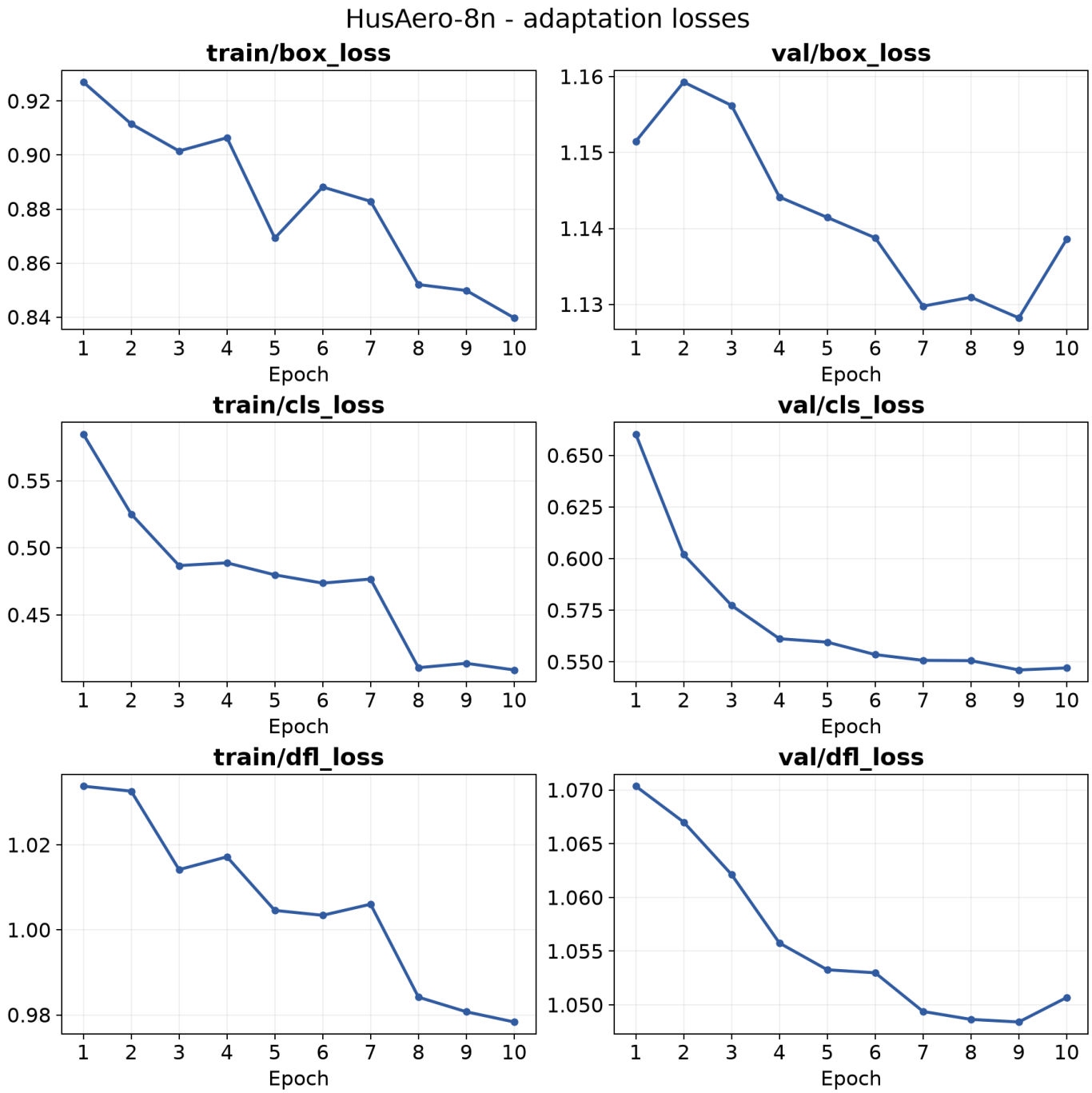
All 10 completed adaptation epochs. These are training/validation histories; final test measurements are reported separately. Best checkpoint: epoch 8.



Numeric source: results/training/adaptation10/HusAero-8n/results.csv. Full settings: configs/adaptation_protocol.json.

04 / HusAero-8n loss curves

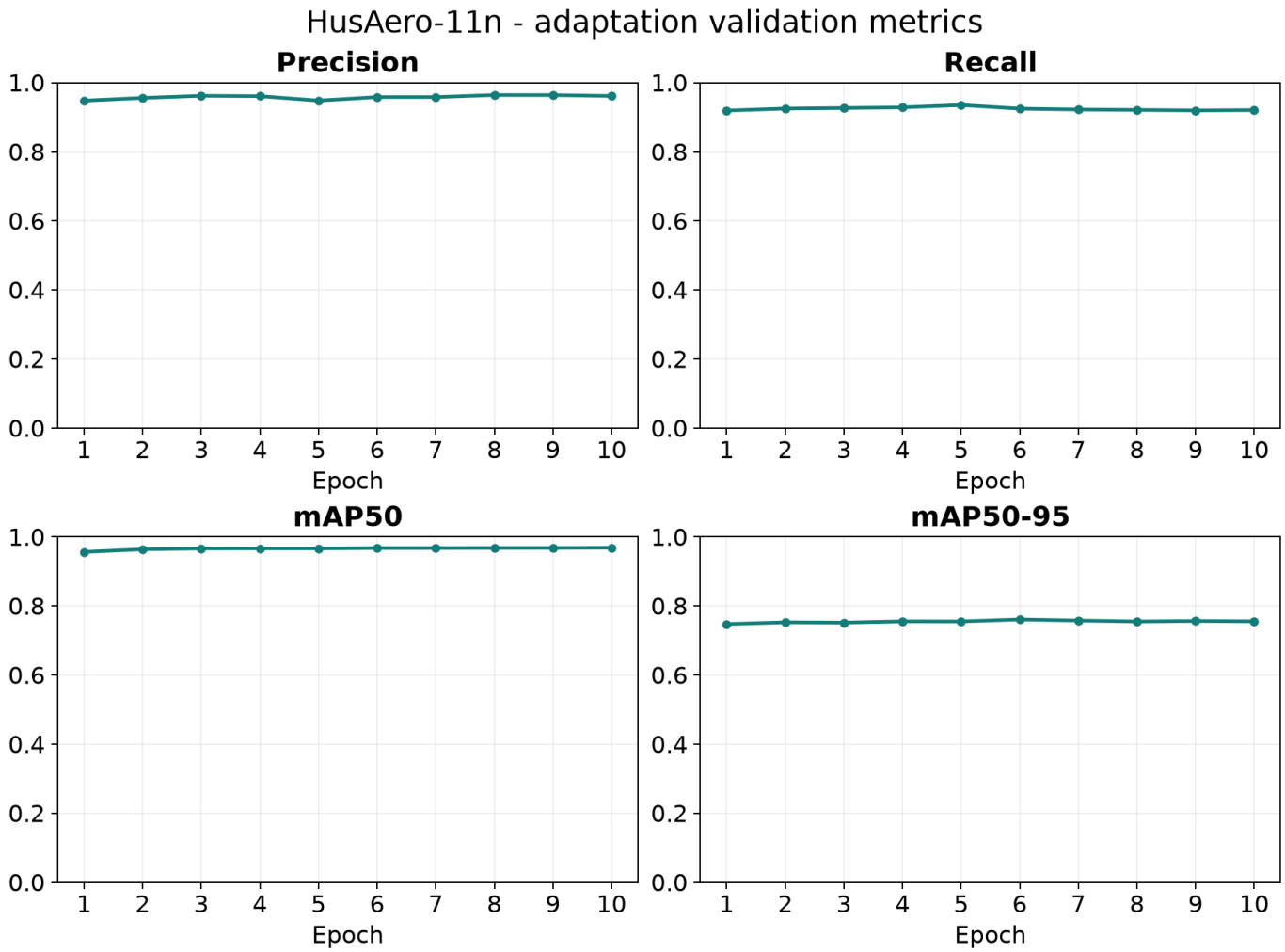
All 10 completed adaptation epochs. These are training/validation histories; final test measurements are reported separately. Best checkpoint: epoch 8.



Numeric source: results/training/adaptation10/HusAero-8n/results.csv. Full settings: configs/adaptation_protocol.json.

05 / HusAero-11n validation metrics

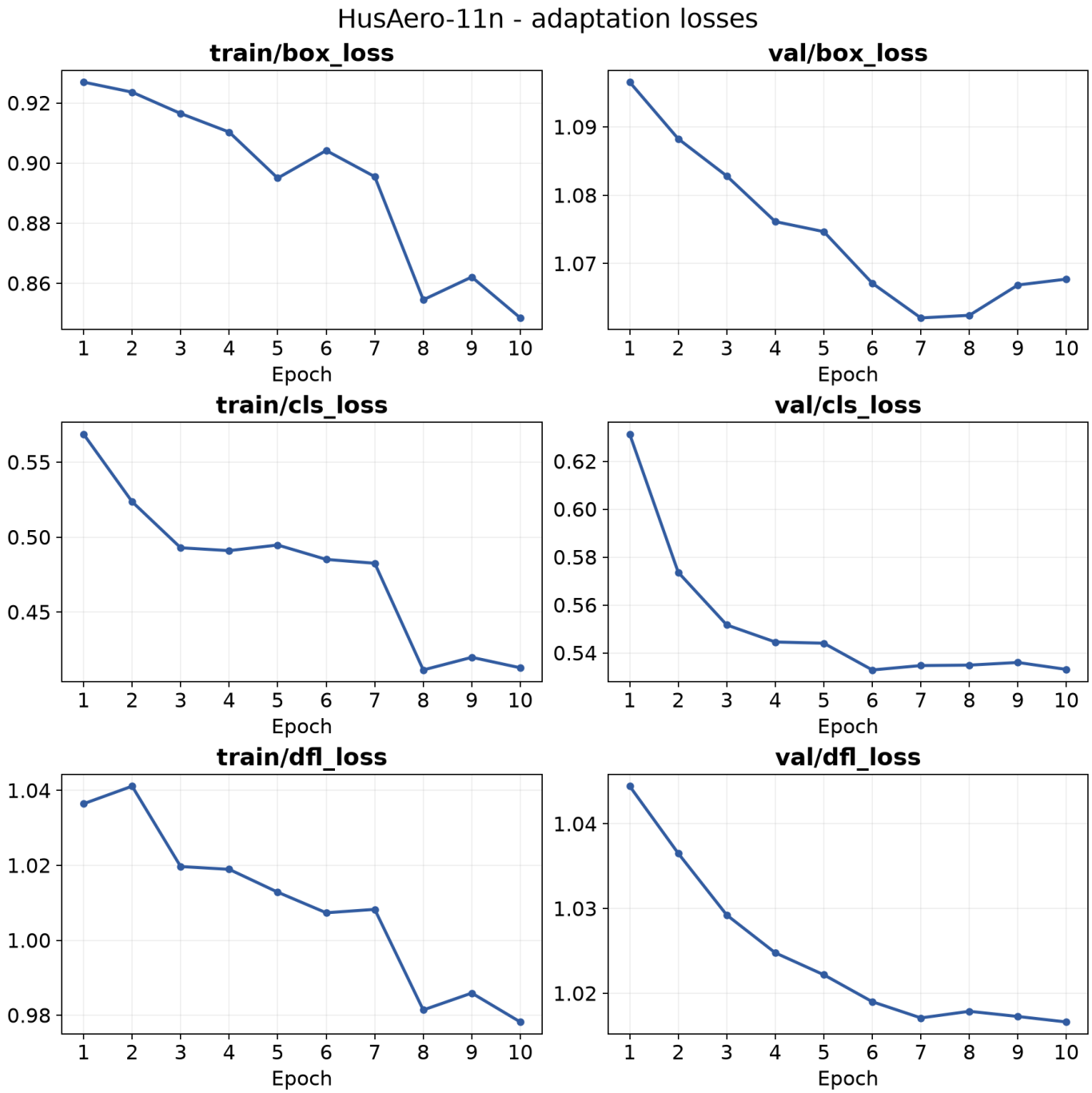
All 10 completed adaptation epochs. These are training/validation histories; final test measurements are reported separately. Best checkpoint: epoch 6.



Numeric source: results/training/adaptation10/HusAero-11n/results.csv. Full settings: configs/adaptation_protocol.json.

05 / HusAero-11n loss curves

All 10 completed adaptation epochs. These are training/validation histories; final test measurements are reported separately. Best checkpoint: epoch 6.



Numeric source: results/training/adaptation10/HusAero-11n/results.csv. Full settings: configs/adaptation_protocol.json.

06 / Actual test-image predictions

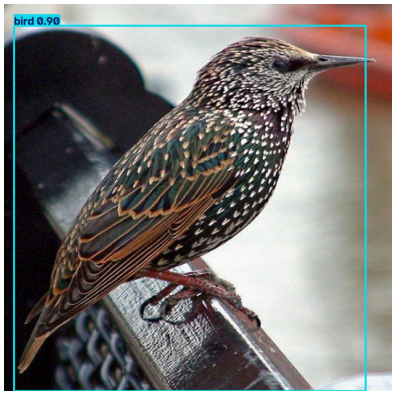
Same public sample images, PC CPU inference, confidence 0.25. Predictions only; no GT overlay. These are examples from the public test split, not separate unseen data.

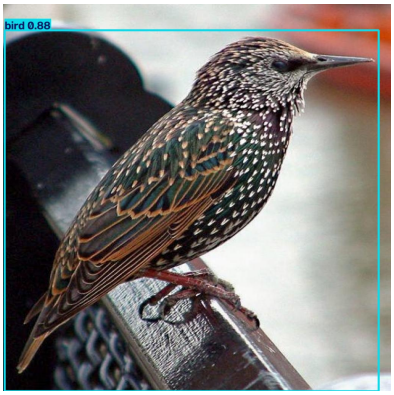
Drone (multirotor) | HusAero-8n (left) / HusAero-11n (right)





Bird | HusAero-8n (left) / HusAero-11n (right)





Airplane | HusAero-8n (left) / HusAero-11n (right)





Photography: MBAOD v1, Firas Mohammed Kadhum, Abbas Abdulazeez Abdulhameed and Jameelah Harbi Saud. Publisher-listed CC BY 4.0. Original watermarks and source credits remain. All 20 paired prediction examples are in results/test_images.

06 / Actual test-image predictions

Same public sample images, PC CPU inference, confidence 0.25. Predictions only; no GT overlay. These are examples from the public test split, not separate unseen data.

Helicopter | HusAero-8n (left) / HusAero-11n (right)





Kite | HusAero-8n (left) / HusAero-11n (right)





Drone (fixed-wing / VTOL) | HusAero-8n (left) / HusAero-11n (right)

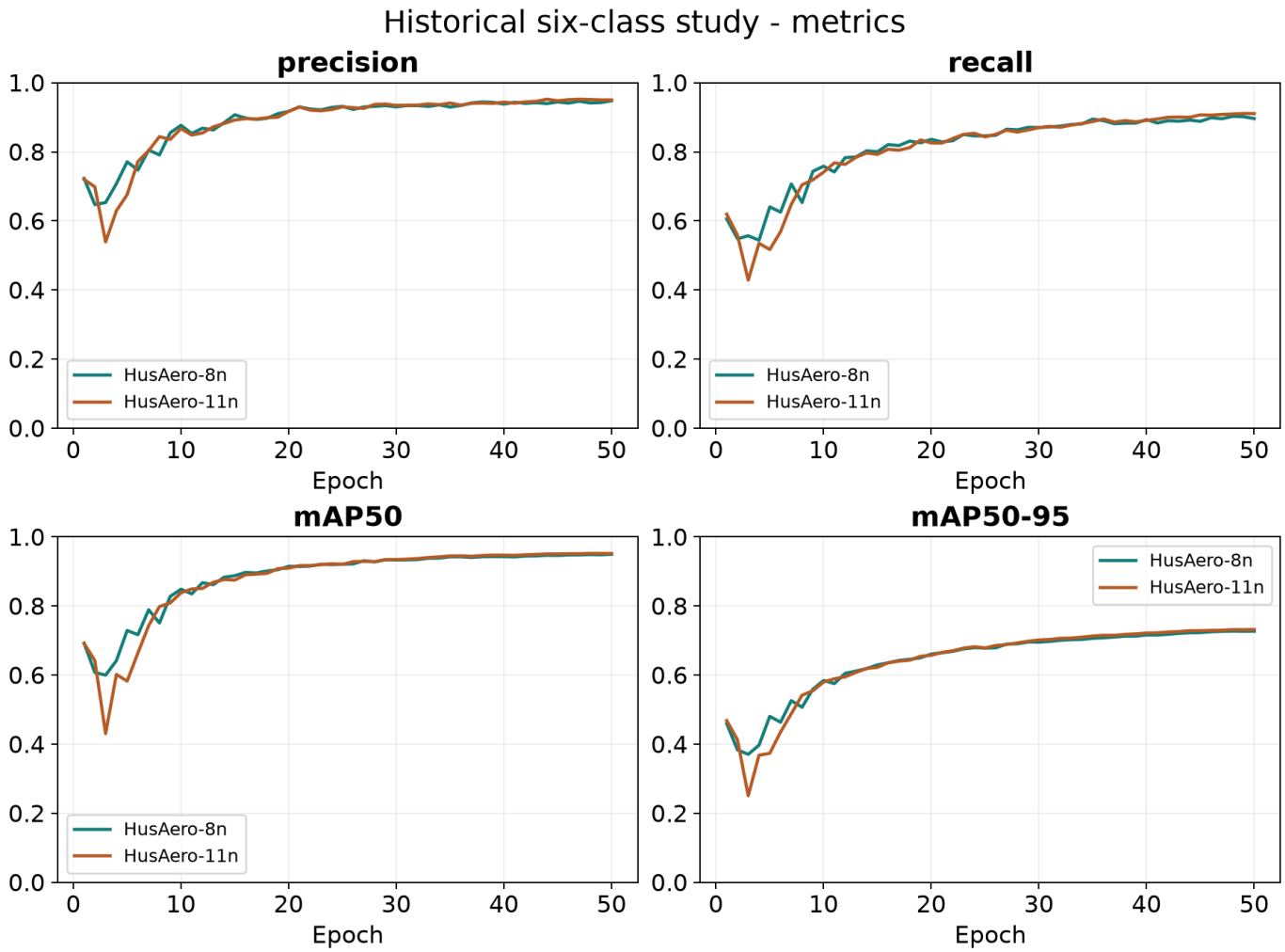




Photography: MBAOD v1, Firas Mohammed Kadhum, Abbas Abdulazeez Abdulhameed and Jameelah Harbi Saud. Publisher-listed CC BY 4.0. Original watermarks and source credits remain. All 20 paired prediction examples are in results/test_images.

A / Historical 50-epoch metrics

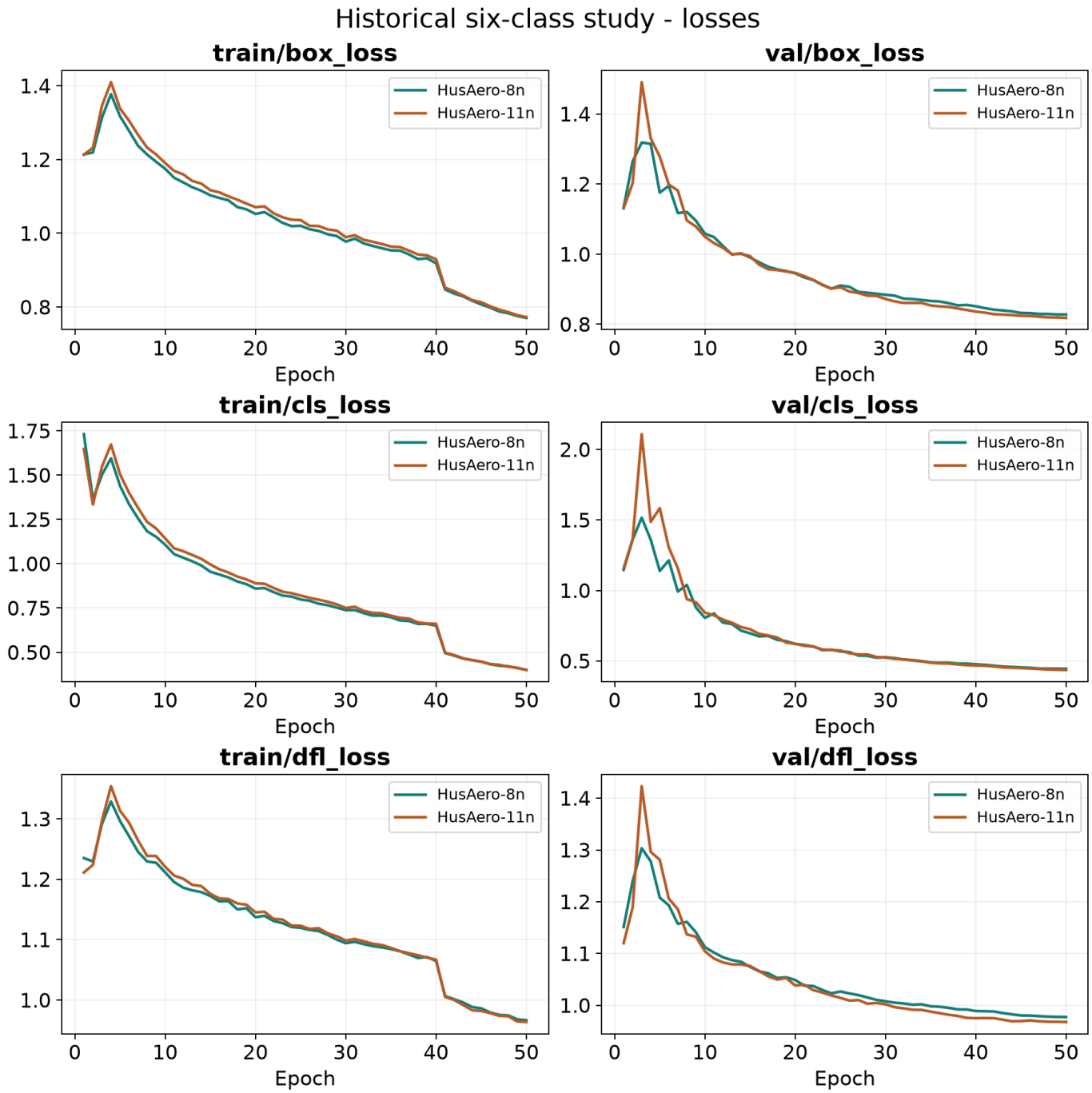
Earlier matched six-class study. Same historical split and training settings for both architectures. These validation curves describe the earlier checkpoints, not the final five-class adaptation test. Original CSV values are preserved byte-for-byte.



Numeric source: results/training/historical50/{HusAero-8n,HusAero-11n}/results.csv. Source credits: Ultralytics (github.com/ultralytics/ultralytics); MBAOD v1 (doi.org/10.17632/zfc8hndczp.1). Dataset screening does not establish exhaustive label or session independence.

A / Historical 50-epoch losses

Earlier matched six-class study. Same historical split and training settings for both architectures. These validation curves describe the earlier checkpoints, not the final five-class adaptation test. Original CSV values are preserved byte-for-byte.



Numeric source: results/training/historical50/{HusAero-8n,HusAero-11n}/results.csv. Source credits: Ultralytics (github.com/ultralytics/ultralytics); MBAOD v1 (doi.org/10.17632/zfc8hndczp.1). Dataset screening does not establish exhaustive label or session independence.